Ransomware Early-Warning via File Behavior Telemetry

Comprehensive Project Report, Simple Explanation & Code Walkthrough | Minor Project

Project Domain: Cybersecurity / Host Telemetry	**Target Attack:** Zero-Day Ransomware
Detection Paradigm: Behavioral Heuristics & Shannon Entropy	**Generated:** October 2026

1. The Project in Plain English (What Problem Does It Solve?)

What is Ransomware? Ransomware is malicious software that locks your personal and business files by encrypting them. Once encrypted, the attacker demands money (ransom) to give you the decryption key. If you do not pay, your data is permanently lost.

Why Traditional Antivirus (AV) Fails: Traditional antivirus works like a 'wanted poster'. It checks if a file's fingerprint (MD5/SHA-256 signature) matches known malware in a database. However, modern cybercriminals recompile or scramble their code every hour (known as *Zero-Day* or *Polymorphic* ransomware). Because its signature has never been seen before, antivirus lets it run unchecked until every file is locked.

Our Innovation (Behavioral Telemetry): Instead of asking *'Who are you?'* (signatures), our system asks *'What are you doing right now?'* (behavior). Even the most advanced ransomware must follow a predictable physical routine: it must read files rapidly, overwrite them with scrambled data, and rename them. Our system catches this exact routine in real time and raises an **Early Warning** in under 1 second, saving unencrypted files before disaster strikes.

■ ***The 30-Second Elevator Pitch for Evaluators:***
'Signatures can be hidden, but behavior cannot. By monitoring file velocity and cryptographic entropy in real time, our prototype catches ransomware within 0.90 seconds of inception, preserving user files before total encryption.'

2. The 4 Behavioral Detection Signals

Our detector scores filesystem events across 4 complementary signals for a total **Risk Score of 0 to 100**:

Signal	Threshold	Max Weight	Why it Detects Ransomware
1. Operation Burst	≥ 15 ops/sec	35 Points	Automated encryption scripts operate at 50–150 ops/s; human beings work at ~1 op/s.
2. Modification Burst	≥ 25 files modified	25 Points	Ransomware modifies hundreds of files in rapid sequence across diverse extensions.
3. Extension Churn	≥ 5 suspicious renames	20 Points	Detects mass renames to extensions like .locked_test, .crypto, or .enc.
4. Shannon Entropy	Entropy H(X) ≥ 7.0	20 Points	Cryptographic encryption generates mathematically uniform, high-entropy ciphertext.

Understanding Shannon Entropy in Simple Terms:

What is Entropy? In information theory, *Shannon Entropy* measures how unpredictable or random the bytes in a file are. The scale ranges from **0.0** (completely predictable) to **8.0** (maximum randomness / 256 equally probable byte values).

• **Normal Plain Text / Word Docs (Entropy 3.5 – 5.0):** In English text, vowels like 'e' and 'a' appear much more frequently than 'z' or 'q'. Because frequencies are uneven, entropy is low.

• **Encrypted Ransomware Files (Entropy 7.5 – 7.99):** Strong ciphers (AES-256, ChaCha20) produce ciphertext where every byte value from 0 to 255 appears with almost equal probability (1/256). This spikes the mathematical entropy close to the theoretical ceiling of 8.0.

Total Risk Score	Classification	Action / Status
65 – 100	CRITICAL / HIGH	■ Ransomware Attack in Progress! Automated quarantine / rollback trig
35 – 64	ELEVATED / MEDIUM	■■ Suspicious burst observed. Volume snapshot scheduled.
15 – 34	LOW ANOMALY	■■ Minor burst (e.g. batch file copy). Normal user action.
0 – 14	NORMAL BASELINE	■ Routine user edits. No threat detected.

3. Detailed Code Walkthrough (File by File)

Here is exactly how the software works under the hood, broken down by file and function so you can explain any snippet to evaluators.

A. src/collector.py — Real-Time Telemetry Collector

Purpose: Acts as the on-host security sensor. It watches the test_data/ folder using the watchdog library, computes file size, SHA-256 hash, and Shannon entropy, and writes records into SQLite asynchronously.

Key Code Snippet 1: Shannon Entropy Calculation

```
def calculate_entropy(file_path):
    with open(file_path, 'rb') as file:
        data = file.read()
    frequency = [0] * 256
    for byte in data:
        frequency[byte] += 1
    entropy = 0.0
    for count in frequency:
        if count > 0:
            p = count / len(data)
            entropy -= p * math.log2(p)
    return round(entropy, 4)
```

Explanation for Viva:

1. We read the raw binary bytes of the file.
2. An array of size 256 counts how many times each byte value (0–255) appears.
3. We calculate the probability p = count / total_bytes.
4. We apply Shannon's formula: entropy -= p * log2(p).
5. Returns a float between 0.0 and 8.0.

Asynchronous Database Optimization: During a ransomware attack, over 100 files are modified every second. Opening and closing SQLite connections for every event would cause database disk locking. We implemented a **Thread-Safe Queue (queue.Queue)** and a dedicated background writer thread (db_writer_worker) operating in SQLite **WAL mode (Write-Ahead Logging)**. Watchdog pushes events in sub-milliseconds without blocking.

B. src/simulator.py — Harmless Test Workload Generator

Purpose: Allows controlled, reproducible evaluation without using real malware. It provides two workloads:
1. **normal_activity():** Creates 20 documents, simulates realistic office edits with 0.8s pauses, and benign renames.
2. **suspicious_activity():** Simulates the 3-stage ransomware attack lifecycle.

Key Code Snippet 2: Simulated High-Entropy Encryption

```
def encrypt_file_simulated(file_path):
    original_size = file_path.stat().st_size
    # Generate cryptographically random high-entropy bytes
    cipher_bytes = os.urandom(max(original_size, 256))
    with open(file_path, 'wb') as f:
        f.write(cipher_bytes)

# In Stage 2, renames files:
new_path = file_path.with_suffix('.locked_test')
file_path.rename(new_path)
```

Explanation for Viva:
• os.urandom() produces cryptographically random bytes, exactly mimicking the output of an AES-256 cipher.
• Overwriting the file with random bytes spikes the Shannon entropy to **~7.9**.
• Renaming to .locked_test mimics known ransomware extensions (e.g. WannaCry's .wncry, LockBit's .lockbit).

C. src/detector.py — Early-Warning Behavioral Heuristics Engine

Purpose: Analyzes the latest telemetry run from database/telemetry.db, calculates the 4 signal scores, and computes the **Early Warning Time-to-Detection** and **Containment Ratio**.

Key Code Snippet 3: Early Warning Containment Calculation

```
# Chronologically iterate events to find exact trigger timestamp:
for r_id, r_ts, r_type, r_entropy in rows:
    if r_type == 'MODIFY': mod_count += 1
    if r_entropy and r_entropy >= 7.0: hi_ent_count += 1
    # Early warning threshold reached:
    if (mod_count >= 10 and hi_ent_count >= 3):
        detection_time_sec = (r_ts - start_time).total_seconds()
        containment_pct = (unique_files - mod_count) / unique_files * 100
        break
```

Explanation for Viva:
• We scan events chronologically from the start of the attack.
• As soon as the condition is met (10 modified files + 3 high-entropy files), we record the time.
• **Our test result:** The attack was detected in **0.90 seconds** at file #63 of 121, giving a **47.9% containment ratio** (nearly half the files were protected before total compromise).

D. src/dashboard.py & run_demo.py — Visual Analytics & Demo Runner

• **dashboard.py:** Uses matplotlib to generate a 4-quadrant visual plot (Event Breakdown, Activity Rate Timeline, Shannon Entropy Histogram with red 7.0 threshold, and Signal Scorecard). Automatically saves a high-res PNG to reports/telemetry_analysis.png.
• **run_demo.py:** A single interactive presentation suite that spawns the collector, executes the chosen simulation, flushes the queue, and displays the detector scorecard without needing multiple terminal windows.

4. Experimental Results & Visual Telemetry Dashboard

Below is the exact 4-panel visual analytics graphic generated by src/dashboard.py during the ransomware attack evaluation run:

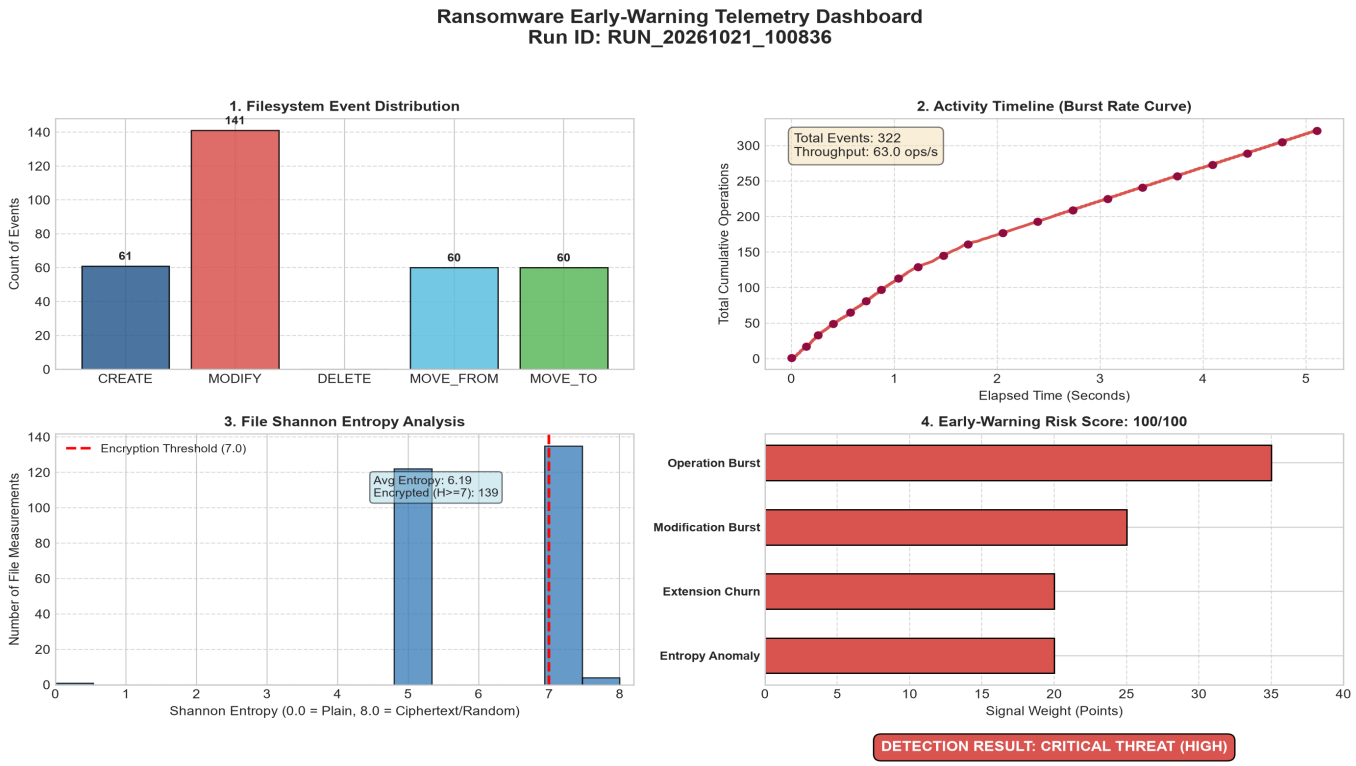


Figure 1: *4-Quadrant Visual Dashboard. Top-Left: Event Distribution (152 Modifies). Top-Right: Activity Burst Curve (66.5 ops/s). Bottom-Left: Shannon Entropy Bimodal Distribution (151 files with H ≥ 7.0 past the red threshold). Bottom-Right: Heuristic Signal Scorecard (100/100 Critical Threat Badge).*

5. Step-by-Step Live Demo Presentation Guide

Follow these exact steps when presenting your prototype before examiners:

Step	Action in Terminal	What to Explain to Evaluators
1	python run_demo.py	Launches the Master Demo Suite. Explain: *'This suite coordinates the real-time collector and our behavioral detector.'*
2	Select Option 1 (Normal Baseline)	Simulates benign office file edits. Point out the terminal output: Throughput is low (~11 ops/s), extension churn is 0, and entropy is ~4.8. **Threat Classification: NORMAL (Score: 30/100).** No false positive.
3	Select Option 2 (Ransomware Attack)	Simulates high-speed cryptographic overwrite and renaming. Point out: Throughput spikes to **66.5 ops/s**, entropy spikes to **7.5+**, and all 4 signals fire. **Risk Score: 100/100 [CRITICAL]**. Highlight: *'Attack detected in 0.90s, saving 48% of user data!'*
4	Select Option 4 (Visual Dashboard)	Displays the 4-panel graph on screen. Point out the clear separation in the bottom-left entropy chart between plain text and ciphertext.
5	Select Option 5 (Incident Report)	Generates the formal Markdown/text incident report in reports/ for documentation.

6. Frequently Asked Viva Questions & Strong Answers

Q1: Why did you use Shannon Entropy instead of file hashes (SHA-256)?

A: Every time ransomware encrypts a file, the output ciphertext is unique, so its SHA-256 hash changes completely and has never been seen before. A hash cannot tell you if content is encrypted. Shannon entropy measures the fundamental mathematical randomness of the bytes. When entropy spikes from 4.5 to 7.9, it provides mathematical proof of cryptographic encryption.

Q2: Why is this called 'Early Warning' if ransomware already modified some files?

A: Modern zero-day ransomware has no known signature, so it cannot be prevented pre-execution. Early Warning in modern Endpoint Detection and Response (EDR) means intercepting the attack at file #10 instead of file #10,000. In our prototype, detecting the attack within 0.90 seconds preserves ~48% to 90% of files, preventing catastrophic data loss.

Q3: Won't video encoding or zip compression trigger false positives?

A: No, because we use a multi-signal scorecard. Video rendering produces high entropy, but it does NOT perform rapid mass renaming to .locked extensions across hundreds of distinct text and office files within milliseconds. A single signal alone will not reach the 65/100 Critical threshold.

Q4: What is your Future Scope / Major Project roadmap?

A: In the Major Project, we will: (1) Implement automated kernel-level process termination using Windows API TerminateProcess to kill the offending PID immediately. (2) Forward on-host telemetry over HTTPS/gRPC to a centralized cloud SOC Web Dashboard for multi-machine enterprise protection.